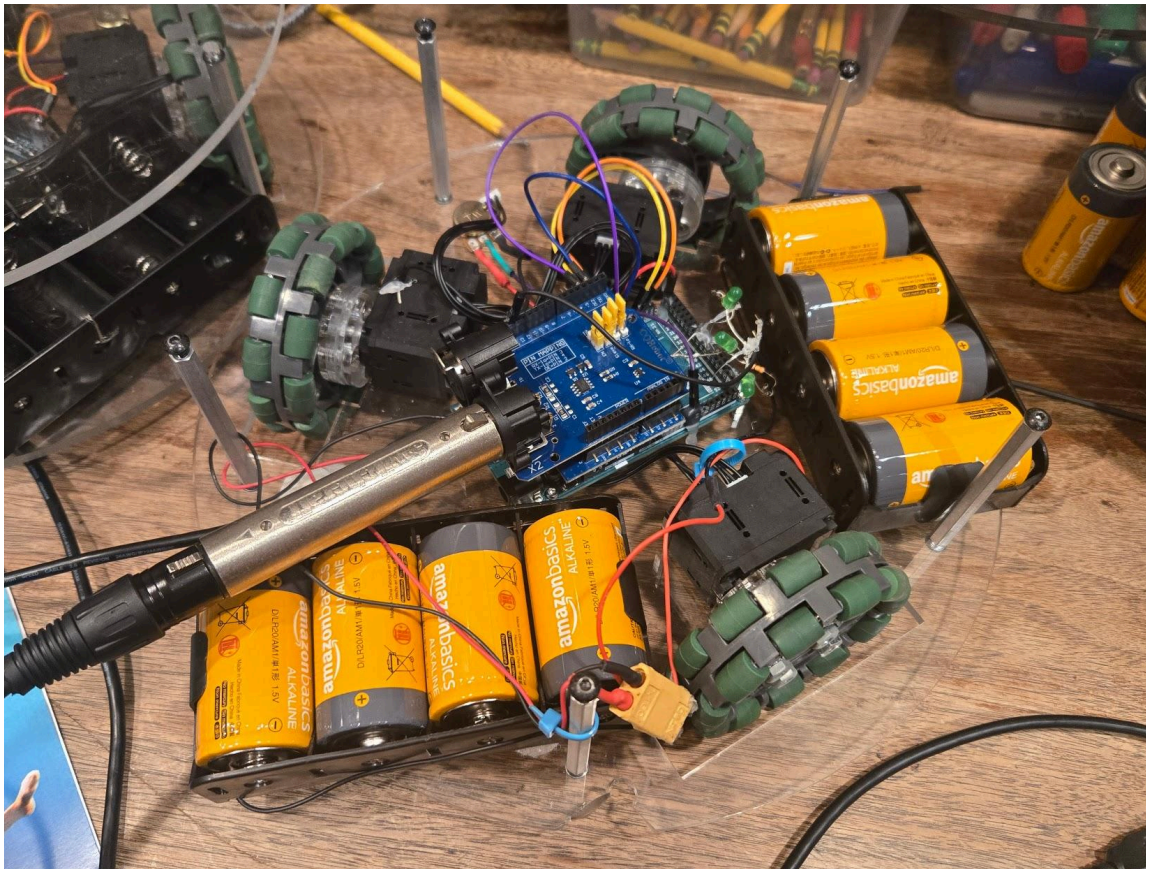
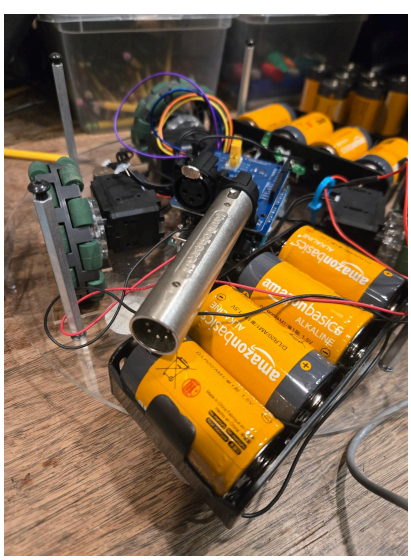
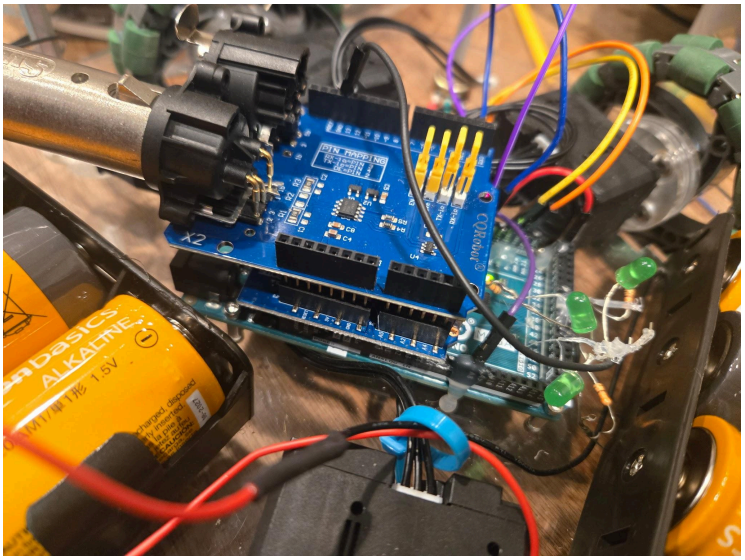
Project Overview

For an AW show a couple years ago, Dr. J and I built a couple robots that would run certain programs to drive across the stage, with little 3D printed rhinos on them. The bots were circular, with 3 omni wheels driven by 3 dynamixel motors, all controlled by an arduino with a dynamixel shield on top, powered by 8 D batteries. I made this bot controlled through DMX wirelessly, with instructions below. Here are some pictures so you understand what I was doing.



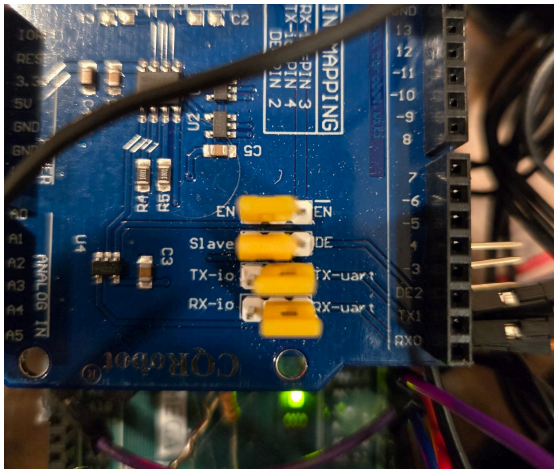




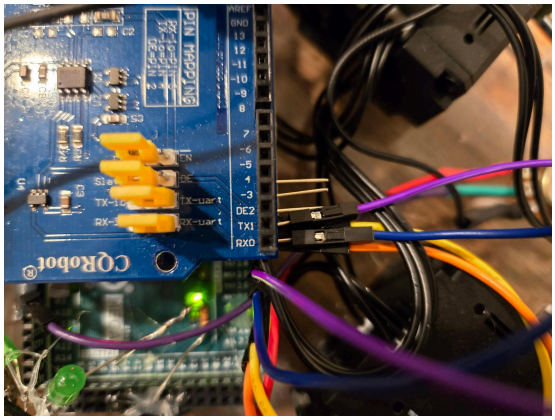
Here’s an explanation and guide of what I did to set this up

1. Setting up the shield

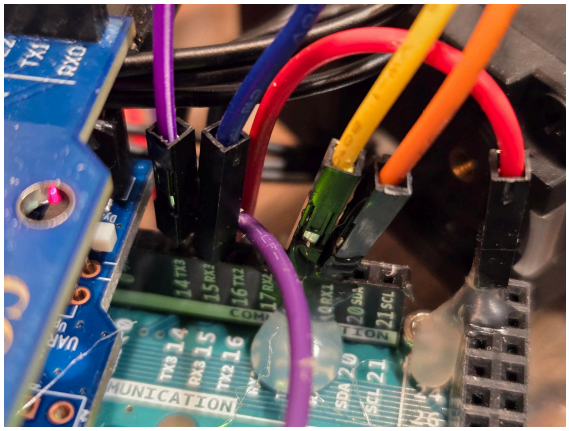
We used an Arduino Mega, mostly because that was what was already on the bot, but also because it had multiple sets of Tx and Rx pins, which was necessary because the dynamixel controller shield and the DMX shield both use them for communication.



For using the arduino as a slave, this is how you want the yellow jumpers configured.



For both the dynamixel and the DMX shield, we bent up the Tx and Rx pins and used male-female jumper cables to connect ones on the arduino (see image below). This is because both shields try to communicate through those one set of pins, which would be a problem if they were just plugged into each other as normal. The jumper cables allow us to plug them into different ports.



Here is where they were plugged into the arduino.

2. Libraries and Code

I used this library for communication, mostly because many other libraries focus only on using the arduino as a master rather than a slave. Here is a repository with an example file that hopefully contains all the info and code you need to apply this to any project, as well as the final code I used specifically for the rhino bots. The only other thing you would have to change is line 70 of Conceptionics.h; This controls what serial ports the shield uses. If you have it set up like mine, with the Tx and Rx plugged into 14 and 15, you will have to change the number to 3. There's not much else to note that isn't explained in the example file.

3. DMX

To control these from your computer, you need a ETCNomad key and a Gadget II plugged into your computer. The Nomad key just gives you permission to send commands, and the Gadget is where they are outputted. The showbabies were perhaps the easiest part; just plug one into the first DMX port on the gadget and the other goes on the bot. Because the shield is 3 pin, you will need a 3 pin female to 5 pin male adapter as shown in the first couple images. Then just plug the showbaby into the shield. If you are using the light board, you can just plug the first showbaby into the board directly.

4. Creating a Fixture

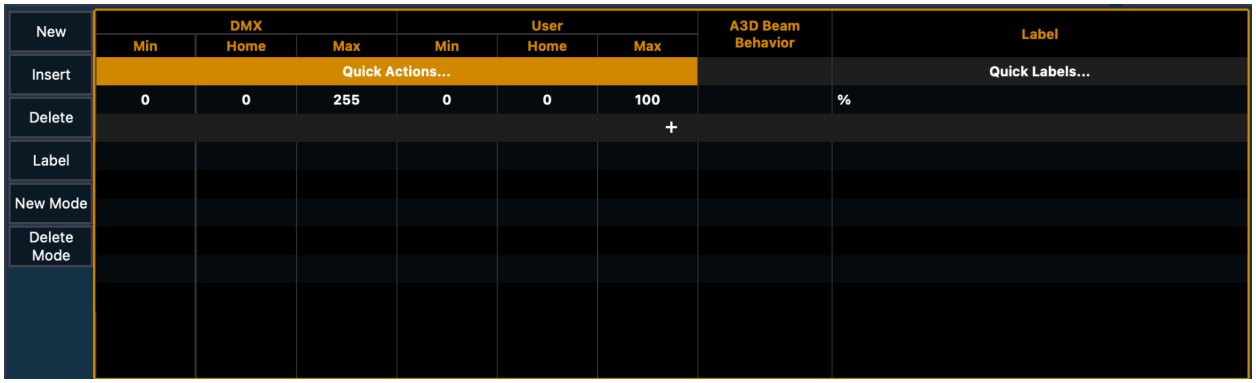
In the Patch tab, select the channel you want to use, and then Type. Click the Edit button at the bottom of the screen.



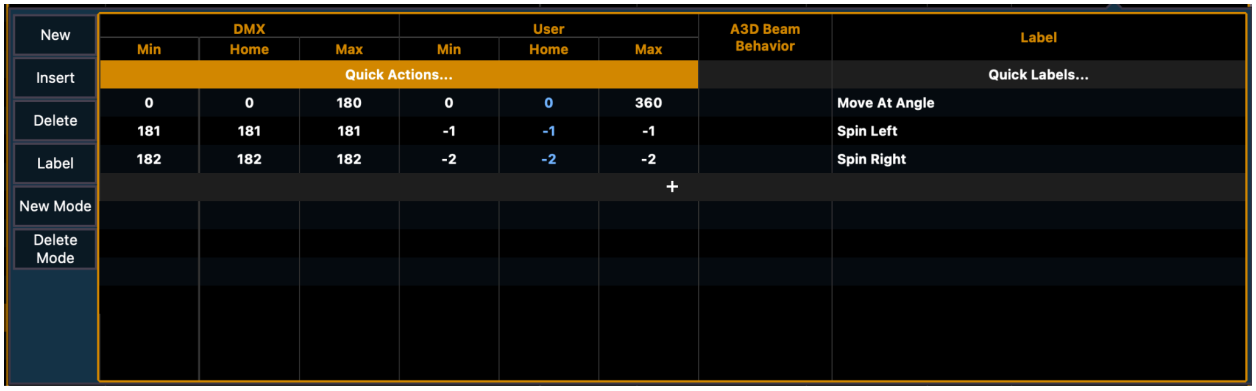
This should bring you to a screen that looks like the one below. To create a new fixture, click New in the bottom left corner.



Here you can add and edit parameters of your fixture. The parameter name can be chosen from the many premade ones. Note that if you do, for example, “Channel 70 @ Full” without specifying the parameter, it will always change the intensity. There is no way to change that. If you do not have an intensity parameter, it will do nothing. The DMX column specifies the data size (here, 8bit) and the position the parameter takes in the DMX addresses. So here, if we set our start address as 190, Intensity would be address 191. Snap is whether, when the parameter is changed between cues, it should fade the value to the new one (as you would for light intensity), or simply set it with no fade time. With my macro parameter, which controls the program it's running, fading as you would light intensity would be problematic. Therefore, I checked the Snap box, so it wouldn’t do that. I did run into some problems when exiting out of an effect where it would fade, so be wary of that.



The last thing is ranges. Most simple parameters, like the intensity of a normal light as displayed, will only have one range. The DMX values are what the DMX actually outputs, and the user values are what is displayed and edited on the light board. For example, on the board you can set this light to a brightness 0-100%, with 0 as the default. However, the value the DMX outputs is scaled from its min and max, so it outputs from 0-255. Thus if you do “Channel 1 @ 50”, the value the DMX outputs is 128. The home is the default value. Usually you want DMX home and User home to correspond. If you were, here, to change the DMX home to 128, but keep the User home 0, clicking the home button would show the user 0 but output 128, while explicitly setting the fixture to the home value “Channel 1 @ 0” It would still do the normal scaling math and output 0. Probably don’t mismatch them unless you have a good reason to.



These are the ranges for my Macro parameter. Having multiple ranges like this allows them to show up in the ML controls as separate options, as shown below. This makes it a lot more clear and easy to use, so your user doesn't have to memorize what “program 6” is. It also allows me to have values the user can set that you don't necessarily want to scale with the DMX value linearly, as I did.



You can also create “Modes” in this panel, which then has their own set of ranges. Basically you independently set the mode (for my fixture the best analogy would be a degrees mode and a radians mode), and then have different ranges depending on the mode (ex I would have 2 ranges with the same DMX range but the user input range on one would be 0-360 and the other it would be 6.2) . I won’t go too deep into that here, but if you are curious I’d recommend poking around a fixture like “Platinum Beam 5R Extreme Extended” (our 411 movers) which does a ton of stuff with ranges and modes and ML controls I don’t understand.

Now you’re done! Save your fixture, patch your device to a new channel and address, and start writing cues.

Sources/Helpful links/Products

https://learn.adafruit.com/intro-to-dmx/overview
https://sourceforge.net/p/dmxlibraryforar/wiki/DMX%20Slave/
https://www.amazon.com/CQRobot-network-Management-Extended-Functions/dp/B01DUHZAT0
https://www.amazon.com/City-Theatrical-5900-SHoW-Baby/dp/B07STCFHKX